



POS SYSTEM

Kenyan Supermarket

Modern Point-of-Sale Solution for East African Retail

NestJS

React

Electron

PostgreSQL

M-Pesa

Barcode

The Problem

Kenyan supermarkets face critical operational challenges



Manual Processes

Handwritten receipts, manual stock counts, paper-based records leading to errors and lost revenue



No Real-Time Data

Owners have zero visibility into sales, stock levels, or cash flow during operating hours



Payment Friction

Separate M-Pesa processes, manual reconciliation, long queues at checkout

Kenya loses an estimated KES 2.5 billion annually to retail inventory shrinkage and poor stock management

Our Solution

A complete, modern POS system built for Kenyan supermarkets

- ✓ **Barcode scanning** — Instant product lookup, zero typing errors
- ✓ **M-Pesa STK Push** — Seamless mobile payments at the till
- ✓ **Real-time inventory** — Know your stock levels every second
- ✓ **Cash management** — Automated float tracking and reconciliation
- ✓ **Department reports** — Print performance reports by category
- ✓ **Multi-user roles** — Admin, Manager, Cashier, Inventory Clerk

Built for Kenya

- ✓ KES currency with 16% VAT (KRA compliant)
- ✓ M-Pesa Lipa Na M-Pesa integration
- ✓ Works offline — sync when connected
- ✓ Thermal receipt printer support
- ✓ Runs on any Windows/Mac/Linux PC

Key Features



POS Terminal

Barcode scanner integration, auto-focus input, instant cart management



Inventory

Stock tracking, low-stock alerts, inter-branch transfers, damage logging



Cash Management

Session tracking, cash in/out, expense recording, end-of-day reconciliation



Reports

Department reports, sales analytics, top products, printable PDFs



Customers

Loyalty program, Bronze/Silver/Gold tiers, point tracking



M-Pesa

STK Push payments, automatic callback, real-time confirmation



Security

Role-based access, JWT auth, PIN login, full audit trail



Desktop App

Electron packaging, runs offline, auto-updates ready

Technical Architecture

Backend (NestJS)

- ✓ 15 modular API controllers
- ✓ 60+ RESTful API endpoints
- ✓ Prisma ORM with PostgreSQL
- ✓ JWT authentication + RBAC
- ✓ Swagger API documentation

Database (PostgreSQL)

- ✓ 30+ relational tables
- ✓ 12 enum types
- ✓ Full ACID compliance
- ✓ Cloud or local deployment

Frontend (React + Electron)

- ✓ 9 interactive pages
- ✓ Tailwind CSS design system
- ✓ Zustand state management
- ✓ Recharts data visualization
- ✓ Offline-first architecture

Integrations

- ✓ Safaricom M-Pesa API
- ✓ USB barcode scanners (HID)
- ✓ Thermal receipt printers
- ✓ PDF receipt generation

API Endpoints — 60+ Routes

Module	Endpoints	Key Functions
Auth	1	JWT login with PIN verification
Products	6	CRUD, barcode scan, categories, brands
Sales	7	Create, list, receipts, refunds, daily summary
Payments	3	M-Pesa STK Push, status check, history
Customers	7	CRUD, loyalty balance, history, redeem
Inventory	5	Stock levels, low stock, adjust, transfer, damage
Cash Management	6	Open/close sessions, cash in/out, petty cash
Reporting	7	Daily summary, top products, department, KPIs
Settings	4	System config, M-Pesa credentials, terminal
Promotions	5	Discount rules, BOGO, active promotions
Sync	5	Offline sync push/pull, queue management
Employees	4	Staff management, shifts, schedules
Suppliers	6	Suppliers, purchase orders, receive goods
Audit	2	Activity logs, compliance tracking

Database Schema

Core

- ✓ Users & Roles
- ✓ Permissions
- ✓ Branches
- ✓ Terminals

Products

- ✓ Products & Variants
- ✓ Categories & Brands
- ✓ Stock Movements

Transactions

- ✓ Sales & Sale Items
- ✓ Payments
- ✓ Refunds

Financial

- ✓ Cash Sessions
- ✓ Cash Movements
- ✓ Expenses

Operations

- ✓ Inventory Levels
- ✓ Suppliers
- ✓ Purchase Orders

Analytics

- ✓ Audit Logs
- ✓ Loyalty Transactions
- ✓ Sales Forecasts

30+ Tables

12 Enums

Full ACID

UUID Primary Keys

User Roles & Permissions

Feature	Super Admin	Manager	Cashier	Inventory Clerk
Create/View Sales	✓	✓	✓	—
Process Refunds	✓	✓	—	—
Manage Products	✓	✓	View	✓
Inventory Adjust	✓	✓	—	✓
Customer Management	✓	✓	Create	—
View Reports	✓	✓	—	—
Cash Sessions	✓	✓	Open	—
User Management	✓	—	—	—

Security: JWT token authentication, PIN-based login, full audit trail on every action

M-Pesa Integration

How STK Push Works

- **1. Cashier selects M-Pesa**
Payment method at checkout
- **2. Customer enters phone number**
Safaricom registered number
- **3. STK Push sent**
Prompt appears on customer's phone
- **4. Customer enters M-Pesa PIN**
Confirms payment on their device
- **5. Callback received**
System confirms payment instantly
- **6. Receipt generated**
Sale completed, stock updated

Why M-Pesa Matters

- ✓ 73% of Kenya's GDP flows through M-Pesa
- ✓ 33+ million active users in Kenya
- ✓ Average supermarket: 35% M-Pesa payments
- ✓ Zero cash handling risk for large amounts

Technical Details

- ✓ Lipa Na M-Pesa API integration
- ✓ Idempotent callback handling
- ✓ Automatic retry on timeout
- ✓ Transaction ID verification

Why Choose Us



Local-First

Built for Kenya from day one. KES, VAT 16%, KRA compliant, M-Pesa native



Offline-Capable

Works without internet. Syncs when connected. Never loses a sale.



Affordable

Fraction of the cost of SAP, Oracle Retail, or imported POS systems

Feature	Our POS	Imported Solutions	Manual/Paper
M-Pesa Integration	Native ✓	Add-on (\$\$\$)	✗
Offline Mode	Built-in ✓	✗	✓
KRA Compliance	Built-in ✓	Extra config	✗
Barcode Scanner	✓	✓	✗
Real-Time Reports	✓	✓	✗
Customizable	Full access ✓	Limited	✗

Market Opportunity

12,000+

Supermarkets in Kenya

**KES
2.5T**

Annual Retail Market

73%

Payments via M-Pesa

40%

Still Using Manual Systems

Target Customers

- ✓ Supermarket chains (Naivas, Carrefour, Quickmart)
- ✓ Independent supermarkets
- ✓ Wholesale/retail hybrid stores
- ✓ Chain pharmacies

Expansion Plan

- ✓ Phase 1: Kenya (Nairobi + major towns)
- ✓ Phase 2: Uganda, Tanzania, Rwanda
- ✓ Phase 3: West Africa (Nigeria, Ghana)
- ✓ SaaS model for recurring revenue



Ready to Transform Your Supermarket?

Start processing sales in minutes, not months.
One system. Every payment. Complete visibility.

GitHub

github.com/joshkirui/supermarket-pos

Documentation

[POS_System_User_Guide.pdf](#)